

Illini Portal Fit Engine

A transfer-portal recruiting target board for Illinois Men's Basketball

Author: Mannat Talati · **Built for:** Illinois Men's Basketball Analytics Internship **Live app:** (*Streamlit link — see README*) · **Code:** (*GitHub link — see README*)

1. The problem

Roster construction in college basketball now runs through the transfer portal. Every spring, 1,500+ Division I players enter, and a staff has a few weeks to decide who to pursue. Illinois under Brad Underwood has been one of the most aggressive portal and international programs in the country, so this is not a side project for the staff — it *is* the roster.

The hard part is not finding *good* players; it is finding good players who **(a)** fill a **specific hole** the roster just opened, **(b)** fit how Illinois actually plays, and **(c)** are **realistic** to land. Doing that by eye across all of Division I is slow and biased toward the names you already know.

This tool turns that triage into a ranked, explainable board. It profiles Illinois's roster, detects what's leaving, and scores every eligible D-I player on a transparent **Fit Score** so the staff can start the portal window with a shortlist instead of a blank page.

2. The data — and how I sourced it

Everything is **public BartTorvik data**, pulled live over plain HTTP (no scraping tricks, no paywalled sources like KenPom):

Source	Endpoint	What it gives
Player advanced stats	<code>getadvstats.php?year=YYYY&csv=</code>	~5,000 players/season, 67 columns: archetype role, usage, efficiency, shooting splits, BPM/OBPM/DBPM, rebounding, steals/blocks, class, height, hometown
Team ratings	<code>YYYY_team_results.json</code>	365 teams: adjusted offense/defense, Barthag power rating, tempo

The player feed ships with **no header row**, so I reverse-engineered the column layout and then **validated it against the data itself** before trusting a single number:

- shooting splits reconcile ($FTM \div FTA = FT\%$, etc.),
- $BPM == OBPM + DBPM$ holds for **100%** of rows,
- $total\ rebounds == offensive + defensive$ per game holds for **100%** of rows.

Columns I could not verify are named `extra_NN` and are **deliberately unused**, so every figure the tool reports is one I can explain on a whiteboard.

A real snapshot is committed to the repo, so the app loads instantly for a reviewer and the results are reproducible; a "Refresh from BartTorvik" button re-pulls live data on demand.

3. How the Fit Score works

For the current (2025-26) season, the engine reads Illinois as: **28-9, Barthag #6, the

2 offense in the country (adjOE 131.8), #20 defense, and — notably — the #289 tempo.**

That last number matters: Illinois is an *elite half-court* team, **not** a run-and-gun team, so the model rewards skill and shot-making over raw transition athleticism. The profile is *derived from data*, not assumed.

It then assumes seniors graduate (editable in the app) — for 2025-26 that's lead guard **Kylan Boswell**, stretch big **Ben Humrichous**, and **AJ Redd** — and measures what walks out the door: a chunk of the team's **playmaking, perimeter defense, and outside shooting**.

Each eligible candidate (non-Illinois, has remaining eligibility, cleared minimum minutes/games) gets a **0-100 Fit Score**, a weighted blend of four parts that are each shown separately:

1. **Production** — proven value: BPM, porpag, minutes load.
2. **Role fit** — does their position fill a *depleted* Illinois spot (guard / wing / big)?
3. **System fit** — do their strengths match the **specific** skills Illinois must replace? The weights here come straight from the departure analysis (e.g. "we lost 24% of our made threes" → shooting is weighted up).
4. **Attainability** — a realism lens from the player's current team strength, surfaced as a plain label: *High-major proven, Mid-major riser, Low-major standout*.

Every component is percentile-ranked **within the candidate pool**, so a score answers the question a coach actually asks: *"compared to the other realistic options, how good is this?"* All four weights, the departure list, and the need weights are **tunable live** in the app.

Example output (default Illinois needs, 2025-26):

Fit	Player	Team	Role	Why
86.8	Jason Rivera-Torres	Monmouth	Wing F	3.3% steal rate; 1.5 3PM/g; +5.5 BPM — perimeter D + shooting; mid-major riser
85.3	Tyler Tanner	Vanderbilt	Scoring PG	5.1 ast/g (29 AST%); +10.2 BPM — playmaking; high-major proven
85.1	Mason Falslev	Utah St.	Combo G	3.6% steal rate; 3.1 ast/g; +8.8 BPM — perimeter D + playmaking

Each row carries an auto-generated scouting note, and the **Player Detail** view breaks the score into its four bars next to the player's full statistical profile.

4. Three tools that make the board a scouting product

A ranked list is a start; a staff also needs to know *what the numbers mean at our level, who a player reminds us of*, and have something they can **print and hand to a coach**. Three additions do that — and the first two are calibrated on real history, not assumed.

4a. Big Ten Translation — what a box line becomes at Illinois's level

A 20-and-8 against low-major defenses is not a 20-and-8 in the Big Ten, because box-score counting stats are **not** opponent-adjusted. To make gaudy lines honest, I learned the deflation from data: I matched the **same player across consecutive seasons** using BartTorvik's stable player id, found everyone who **changed competition level**, and measured how much of each stat survived the jump. The calibration set is **990 real cross-season transfers**.

For each stat I fit a one-parameter retention line, anchored so a zero-gap move means no change ($\text{ratio} = 1 + \text{slope} \times \text{level_gap}$), where the gap is the Barthag distance between the player's team and Illinois (Barthag **0.968**). The result is exactly the scouting lesson you'd hope for:

Stat	Survives a jump?	Keeps, at a typical mid-major→Illinois jump
Points / g	deflates most	~78%
Rebounds / g	deflates	~84%
Assists / g	deflates	~86%
Usage%	deflates mildly	~88%
Made 3s / g	travels well	~90%
TS% (efficiency)	holds (slope ≈ 0)	~101%

So **efficiency travels and made-shooting travels; raw volume deflates** — and the deflation scales with how big a leap the player is making. Concretely, a 21.4 ppg scorer in the SWAC projects to **~6.5 ppg** in the Big Ten ("major jump, high variance"), while a high-major star at BYU keeps ~93% of his scoring. The board shows each candidate's **raw line next to its projection**, a **Keep%**, and a **level-jump risk label** so a 25-ppg low-major name can't masquerade as a 25-ppg Big Ten name.

Note on double-counting: the Fit Score's *production* component already uses opponent-**adjusted** inputs (BPM, adjusted ratings), so the translation model intentionally leaves the score alone and instead translates the **raw box stats coaches read directly** — keeping both numbers honest and separate.

4b. Player comps — "who does he play like?"

During portal season a staff constantly asks two questions: *a starter just left — which available transfers replace his style?* and *we like this target — who does he remind us of?* Both are nearest-neighbour problems in a **style space**: each player becomes a vector of role/skill rates (usage, shot diet, playmaking, ball security, rebounding, rim protection, perimeter activity, size, efficiency), **z-scored against the D-I rotation baseline** so every axis is comparable. Distance is plain Euclidean — no learned weights — reported as a 0-100 similarity. The **Program & Needs** page turns a departing Illini into a ranked list of transfer-eligible look-alikes; every target carries its closest high-major comps.

4c. Visual scouting cards + printable PDF

Coaches read cards, not 60-column CSVs. Each target renders a one-page card: a **percentile radar vs the Big Ten rotation** (every spoke is "how would this skill rank in our league?"), the **raw → Big Ten projected** box line, a **shot diet** (rim / mid / three share and accuracy), and the player's **comps** — exportable to a **self-contained 1-page PDF** the staff can print for the board.

5. Four more tools: from a ranked list to a war room

The three tools above make the board *readable*; these four make it *decisive* — they answer the follow-up questions a staff asks the moment a name rises to the top. The first two also turn the model's honest limitations into strengths, and all four reuse the same public, cross-season BartTorvik data (the same player matched across seasons by stable `pid`).

5a. Outcome-backed projections — real precedent behind every number

A projection a staff can't trace is a projection a staff won't trust. So for any target the engine pulls the **real historical transfers** who started from the most similar profile and made the most similar-size level jump — matched in the same z-scored style space, penalised for a mismatch in jump size — and reports **what actually happened to their scoring** the next season. A 21.7-ppg ASun scorer jumping to Illinois surfaces names like Frankie Fidler (20.1 → 7.0), Chaz Lanier (19.6 → 18.0), and — fittingly — Ben Humrichous (14.8 → 7.6), yielding a **band**: "*6 comparable real transfers kept 38–62% of their scoring, median 46% → ~8–14 pts at our level.*" That replaces a single deflated number with an empirical distribution drawn from the same 600+ up-transfers, so the staff sees the *range of outcomes*, not false precision — the named precedent makes the translation defensible in a room.

5b. Development trajectories — who is rising, not just who was good

A single season underrates an ascending player and overrates a one-year spike. By tracking the **same player across seasons** (stable `pid`), the engine measures each returning rotation player's year-over-year move in BPM, usage, efficiency, scoring, and minutes, and labels him **Breakout** (a bigger role that *held* efficiency, still underclass-eligible), **Ascending**, **Steady**, or **Declining**. To keep it honest the trajectory only uses players who held a real rotation role in *both* seasons — a +20 BPM "jump" off a 5-minute freshman year is noise, not growth. The board carries a Δ BPM column and a dedicated **risers** view, and Player Detail shows the full last-year → this-year line — so a sophomore who jumped from −0.8 to +4.0 BPM at a high-major reads as the breakout he is, not as an average-looking single-season line.

5c. Head-to-head comparison — three targets, one screen

When two or three names cluster at the top, the staff compares them directly: overlaid percentile radars (vs the Big Ten), every Fit component, the Big-Ten-projected box line, and shooting/defense rates side by side, each with its one-line scouting note. The decision a coach actually makes — *which of these wings?* — gets its own screen instead of a mental juggle across rows.

5d. Post-portal depth chart — tie every target to a roster hole

Finally, the engine projects the roster **after** the marked departures into a Guard / Wing / Big depth chart against a target rotation (4 / 2 / 3), shows exactly how many spots are open at each position, and lets the staff **slot board targets into the holes** and watch the rotation fill in. It closes the loop from "good player" back to "fills *our* November rotation."

6. How I built it

- **Language:** Python 3.
- **Data / model:** `pandas`, `numpy` — a small, readable package (`illini_fit/`) split into `schema` (validated column map), `fetch` (cache + live refresh), `profile` (team + roster), `needs` (departure-driven gap detection), `fit_score` (the scoring model), `translation` (the Big Ten retention model), `similarity` (the style-comp engine), `precedent` (outcome-backed comparable transfers), `trajectory` (multi-year development), and `scouting` (radar + PDF cards).
- **App:** `streamlit` — an interactive, Illini-themed web app: filters, weight sliders, a ranked board with projected lines and a risers view, a post-portal depth chart, a head-to-head compare screen, player detail with development trajectory + real precedents + scouting card, and CSV + PDF export; `matplotlib` for the radars and `xhtml2pdf` for the printable card.
- **Quality:** every module ships with self-checks run as `python -m illini_fit.<module>` — data row-count/Illinois-parse assertions; a test that a shooting-weighted board surfaces more shooters and scores stay in 0-100; that scoring deflates and efficiency holds in the translation model; that a player is his own closest comp; that the precedent cohort is large and its kept-% band is ordered; that a big BPM jump reads as a riser. The app is verified headlessly with Streamlit's `AppTest`, including the compare and depth-chart interactions.
- **Deploy:** GitHub + Streamlit Community Cloud (one-click, free) for a shareable live link.

7. Why it's useful to a GM / coaching staff

- **It starts the portal window with a shortlist, not a spreadsheet.** Day one of the portal, the staff has a ranked, position-aware board instead of 1,500 names.
- **It's tied to *this* roster's needs.** The board re-ranks the moment you mark a player as leaving — so the night a starter enters the portal, you instantly see who replaces *his* production, not just "good players."

- **It's explainable to non-analysts.** Every score decomposes into production / role / system / attainability with a one-line rationale. That's defensible in a staff meeting and to a head coach who wants the *why*, not a black box.
- **It encodes Illinois's identity from data.** Because system fit is derived from Illinois's actual profile (elite half-court offense, slow tempo, shooting), it won't recommend a player who is good in the abstract but wrong for how Illinois plays.
- **It translates production to *our* level — and shows the receipts.** The Big Ten projection stops a 25-ppg low-major name from masquerading as a 25-ppg Big Ten name, and the precedent engine backs every projection with *real named transfers* who made the same jump and a band of what they actually kept — the single most common mistake in reading raw portal stats, with the evidence attached.
- **It finds risers, not just last year's stars.** Multi-year trajectories flag the ascending sophomore the box score underrates and fade the one-year spike — exactly the edge a development program like Illinois wants.
- **It speaks the staff's language and closes the loop.** "Plays like" comps frame an unknown name in terms of players the room already knows; the compare screen settles *which of these three wings*; the depth chart drops a target into an open rotation spot; and the 1-page PDF card is something a coach can actually print and carry.
- **It's tunable on the fly.** A coach who wants to prioritize a defensive guard over a scoring one just moves a slider; the board updates in real time.

In practice the staff intersects this board with the live portal list (which they have and which is not a clean public feed): the engine ranks the full pool by fit, and the staff filters to who's actually available. That mirrors how an analytics staffer already works — this just does the heavy ranking in seconds.

8. Honest limitations & next steps

- **Portal availability isn't a clean public dataset**, so the engine ranks the full pool and is meant to be filtered to the live portal list. A natural v2 is to ingest a portal feed and auto-filter.
- **The Fit Score still reads one season**, even though the development trajectories now add multi-year context alongside it; a candidate with no prior D-I season (true freshmen, some internationals) has no trajectory, and play-type data (Synergy/Hudl, if licensed) would sharpen role fit further.
- **The translation model is a population average** — which is exactly why the precedent engine now sits next to it: rather than trusting a single deflated number, the staff sees the *band* of what real comparable transfers actually kept. An individual can still beat or miss his comps, so the engine surfaces the range and a level-jump risk label, not false precision.
- **Fit weights are a reasonable default, not gospel** — which is exactly why they're exposed as sliders for the staff to own.

- **No NIL/cost modeling.** A real GM board would layer in budget; that data isn't public.

All data is publicly sourced from BartTorvik. Built as an independent analytics project for the Illinois Men's Basketball Analytics Internship.